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Exploring artificial intelligence applications in construction using a black grey white box approach for predicting project schedule performance in India

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Abstract

Timely completion of construction projects is vital for cost control, stakeholder satisfaction, and project success. In India, however, schedule adherence is frequently challenged by factors such as resource limitations, weather disruptions, and design changes. This study presents an AI-driven predictive modeling framework to forecast schedule performance using real-world data from Indian construction projects. Five machine learning models—multiple linear regression (MLR), support vector regression (SVR), artificial neural networks (ANN), random forest regressor (RFR), and gradient boosting machine (GBM)—were developed and evaluated. Models were categorized as white box, grey box, or black box based on interpretability. MLR, while transparent, showed limited performance in complex scenarios. SVR and RFR offered a balance between accuracy and explainability, whereas ANN and GBM achieved the highest predictive accuracy. GBM emerged as the top performer ($R^2 = 0.93$). Feature importance and sensitivity analyses identified planned duration, workforce size, equipment utilization, and material availability as key drivers of schedule adherence. The findings provide actionable insights for selecting suitable AI models in construction planning and underscore the value of integrating predictive analytics into schedule risk management for Indian infrastructure projects.

Keywords Construction schedule performance, Machine learning models, Black-grey-white box AI, Predictive analytics, Indian infrastructure projects

1 Introduction

India's construction industry plays a critical role in supporting economic development, urbanization, and infrastructure growth [51, 54]. However, one of the persistent challenges in the sector is the inability to adhere to project schedules, often resulting in cost overruns, compromised quality, and stakeholder dissatisfaction [41]. According to government and industry reports, over 35% of infrastructure projects in India experience time overruns due to various factors including delays in approvals, resource constraints,



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and unforeseen site conditions [12, 46]. These delays not only affect project delivery but also have ripple effects on economic productivity and investor confidence [1].

Traditional project scheduling techniques such as the critical path method (CPM), program evaluation and review technique (PERT), and Monte Carlo simulations have long been used to forecast timelines and manage uncertainty [22, 24]. While these methods are valuable, they rely heavily on linear assumptions and predefined probability distributions, limiting their ability to model the complex, dynamic, and nonlinear relationships among scheduling variables [47]. In real-world construction environments, where multiple interdependent factors influence performance, such traditional models often fail to produce reliable predictions [5].

To address these limitations, the construction industry is increasingly turning toward artificial intelligence (AI) and machine learning (ML) techniques for predictive modeling [16, 31]. These data-driven approaches can learn from historical and real-time data to uncover hidden patterns, manage nonlinearity, and improve prediction accuracy [57]. Recent studies have applied various ML models such as multiple linear regression (MLR), support vector regression (SVR), artificial neural networks (ANN), random forest regressor (RFR), and gradient boosting machine (GBM) to tasks such as schedule forecasting, cost estimation, and resource optimization, with promising results [2, 23, 28].

However, one key issue in selecting ML models is the trade-off between predictive accuracy and model interpretability [26]. In high-stakes decision environments like construction project management, stakeholders often require not only accurate predictions but also understandable reasoning behind those predictions [37, 55]. To frame this trade-off, AI models are often categorized as white box, grey box, or black box [36, 53]. White Box models like MLR are fully interpretable and offer transparency, but may underperform when handling nonlinear data [35]. Grey Box models such as SVR and RFR strike a balance, providing partial interpretability with good accuracy [3]. Black Box models like ANN and GBM often achieve the highest accuracy but lack transparency, making their use in critical decision-making more complex [51].

Given this background, the present study aims to develop, evaluate, and compare multiple AI-based predictive models to forecast schedule adherence in Indian construction projects. Using a comprehensive dataset comprising government records, private sector project reports, and real-time monitoring data, this study applies and tunes MLR, SVR, ANN, RFR, and GBM models. The models are evaluated using standard metrics including MAE, RMSE, and R^2 , along with an assessment of interpretability based on the Black-Grey-White Box framework.

The remainder of this paper is structured as follows: Sect. 2 presents a detailed literature review covering traditional scheduling methods, applications of AI in construction, and key factors affecting project schedules. Section 3 outlines the methodology, including data collection, descriptive statistics, correlation analysis, and AI model development. Section 4 discusses the model implementation, hyperparameter tuning, and comparative evaluation of ML models. Section 5 presents the results, including feature importance, sensitivity analysis, and case study validation. Section 6 concludes the study with key findings, practical implications, limitations, and directions for future research.

2 Literature review

Construction scheduling has traditionally relied on deterministic and probabilistic models such as the CPM, PERT, and Monte Carlo simulation [25, 27]. These techniques have long served as foundational tools in project management, helping managers identify critical activities, estimate project duration, and assess risk under uncertainty [10, 43]. CPM and PERT are particularly useful for network-based analysis and task sequencing, while Monte Carlo simulation allows probabilistic risk modeling based on multiple iterations [14, 33]. However, their effectiveness is often limited by assumptions of linearity, static inputs, and inability to incorporate real-time changes or nonlinear dependencies [56]. As construction projects grow in complexity, these traditional models struggle to adapt to the dynamic nature of modern projects and often fall short in handling uncertainty and nonlinear behavior [20].

Recent advancements in artificial intelligence (AI) and machine learning (ML) have opened new avenues for predictive modeling in construction [3, 11, 13, 35]. Unlike traditional methods, AI models are capable of capturing complex, nonlinear relationships among multiple variables [59]. The MLR, though classified as a White Box model for its interpretability, often fails to capture multi-factor interactions in complex environments [34, 52]. In contrast, the SVR and RFR are categorized as Grey Box models that offer moderate transparency and strong predictive capabilities [32]. The SVR, for instance, can handle small-to-moderate datasets efficiently and model nonlinear relationships effectively [58], whereas RFR provides high accuracy with interpretability through feature importance analysis [21].

Black Box models, such as ANN and deep learning (DNN), are particularly effective in identifying complex patterns within large datasets [38, 50]. These models, however, come with trade-offs, including high computational demands and a lack of interpretability, which limits their application in stakeholder-driven decision-making environments [51]. Ensemble models such as GBM and XGBoost have gained attention for their robust predictive performance through boosting techniques [30, 44]. GBM, in particular, has demonstrated high accuracy when hyperparameters are properly tuned, though it remains computationally intensive [39]. XGBoost enhances performance by combining multiple weak learners, making it suitable for tabular prediction tasks [42]. Furthermore, Hybrid AI models that combine two or more algorithms are emerging as effective tools that balance accuracy with interpretability, albeit at the cost of increased model complexity [29].

Numerous studies have explored the influence of various factors on construction schedule performance [9, 15, 18, 40]. These include project size, budget, workforce availability, equipment utilization, contractor experience, resource allocation, and weather conditions [17, 36, 49]. Additionally, factors such as design changes, regulatory delays, and safety incidents can significantly impact timelines and increase uncertainty in project delivery [25]. Recent AI-based studies have incorporated these factors as input variables in predictive models, demonstrating their relevance in improving schedule forecasting accuracy [6, 19]. Notably, ANN and RFR have been successful in identifying complex interdependencies among such factors, which traditional models fail to capture [7, 48]. The key contributions and limitations of previous studies are summarized in Table 1. This comparison highlights the evolution of techniques from deterministic

Table 1 Summary of literature review

Study	Methodology	Key factors	Key findings	Limitations
Jaber et al. [23]	CPM, PERT, Monte Carlo	Cost, Time, Risk	Traditional scheduling methods provide basic forecasts but fail to capture real-time uncertainties	Unable to handle nonlinear project dynamics
Egwim et al. [12]	Multiple Linear Regression (MLR)	Schedule Delay, Budget Variance	MLR provides reasonable accuracy but assumes linearity, limiting prediction capabilities	Struggles with complex, multi-factor interactions
Abioye et al. [1]	Artificial Neural Networks (ANN)	Workforce Productivity, Equipment Utilization	ANN models outperform traditional methods by identifying hidden patterns in large datasets	Requires large datasets and high computational power
Balli & Güven [5]	Support Vector Regression (SVR)	Cost, Sustainability, Risk	SVR models perform well with small-to-moderate datasets and handle nonlinearity effectively	Computationally expensive and requires careful tuning
Ajayi & Chinda [2]	Random Forest Regressor (RFR)	Time, Quality, Safety	RFR demonstrates high accuracy and interpretability for construction schedule predictions	Can be prone to overfitting with excessive trees
Ansari et al. [4]	Gradient Boosting Machine (GBM)	Cost, Time, Environmental Impact	GBM achieves high predictive performance when hyperparameters are optimized	Computationally intensive and requires careful tuning
Černevičienė & Kabašinskas [8]	Decision Tree Regressor (DTR)	Time, Risk, Productivity	DTR provides interpretable results but is highly sensitive to noise	Overfits on small datasets
Ivanova et al. [22]	XGBoost	Cost, Schedule, Performance	XGBoost enhances performance by combining weak learners effectively	Requires fine-tuning for best results
Sahu et al. [46]	Deep Learning (DNN)	Productivity, Resource Allocation	Deep learning models achieve the best accuracy but need large datasets	Lacks interpretability and high training cost
Maurya et al. [34]	Hybrid AI Models	Time, Cost, Material Waste	Hybrid models balance accuracy and interpretability	Implementation complexity is high

models toward AI-based predictive systems, emphasizing the trade-offs between accuracy, interpretability, and computational complexity.

2.1 Gaps in existing research

While AI techniques have shown promise in predicting construction schedules, most studies lack a comparative analysis of multiple models across performance and interpretability. Research often relies on limited or synthetic datasets, reducing real-world applicability, especially in the Indian context. The integration of dynamic, real-time data sources like IoT and BIM remains underutilized. Moreover, the Black-Grey-White Box framework is rarely applied to guide model selection. Key influencing factors such as automation level, regulatory delays, and site accessibility are also understudied. These gaps highlight the need for holistic, explainable, and data-rich AI models tailored to complex construction environments.

3 Methodology

3.1 Data collection and sources

Table 2 summarizes the diverse data sources used to develop the AI-based predictive models. Publicly available government databases (e.g., CPWD, MoHUA, NITI Aayog) provided standardized records on schedules and costs. Private construction firms contributed real-world reports on delays and progress, though access required permission. Research publications and case studies offered insights into influencing factors and model structures. Industry reports from organizations like FICCI and PwC identified

Table 2 Data sources used for model development

Data source	Description	Data type	Availability	Relevance to study
Government Construction Databases	Publicly available project data from Indian government agencies (e.g., CPWD, MoHUA, NITI Aayog)	Project schedules, costs, timelines	Open Access	Provides official records of project performance
Private Construction Firms	Historical project reports from leading construction companies in India	Work progress reports, schedule delays	Limited (Requires permission)	Real-world industry data on scheduling challenges
Research Publications & Case Studies	Published studies on AI-based schedule prediction in construction	Factors influencing schedule performance	Open Access	Provides insights from prior research
Industry Reports (FICCI, PwC, etc.)	Market reports on construction trends and risks	Project cost, risk factors, resource allocation	Subscription-based	Helps in identifying macro-level factors affecting scheduling
Project Management Software Logs (e.g., Primavera, MS Project)	Logs from scheduling software used in Indian construction projects	Task dependencies, resource usage, delays	Requires industry collaboration	Real-time schedule monitoring and performance tracking
Sensor & IoT-Based Monitoring Data	Data from real-time tracking of construction progress (drones, BIM, IoT sensors)	Workforce productivity, equipment utilization, site conditions	Emerging source	Adds dynamic real-time insights into schedule variations

macro-level scheduling risks. Logs from project management software (Primavera, MS Project) supported detailed scheduling data, while sensor and IoT-based monitoring tools, including drones and BIM, supplied emerging, real-time site data. Together, these sources ensured a robust, multi-dimensional dataset reflecting both traditional and modern scheduling influences in the Indian construction context.

3.2 Descriptive statistics and correlation analysis

Table 3 presents the summary statistics of the key input and output variables used in the development of the predictive models. The dataset comprises 150 construction project records and includes a range of project, resource, and risk-related parameters. These descriptive statistics provide insights into the distribution, central tendencies, and variability of the data, which are critical for understanding model behavior and guiding pre-processing tasks such as normalization and outlier handling.

The planned project duration ranged from 182 to 1475 days, with a mean of 785.1 days, indicating a broad spread in project timelines. Similarly, project size varied widely between 666 and 19,345 square meters, reflecting the diversity in project scales. The average project budget stood at ₹224.13 million, while workforce size ranged from as low as 88 to nearly 5000 workers. These variations suggest the need for models capable of handling scale differences across projects.

Key operational factors such as equipment utilization (mean: 65.92%), material availability (mean index: 0.45), and resource allocation efficiency (mean: 71.57%) were included to capture execution efficiency. Human-related parameters like contractor experience, labor productivity, and safety incidents added contextual depth, while risk-related inputs like weather impact, design changes, regulatory delays, and financial risk scores provided insights into uncertainty and schedule vulnerability.

Table 3 Summary statistics of key variables

Variable	Variable type	Count	Mean	Std. dev.	Min	Max
Project duration (planned) (days)	Input	150	785.10	368.82	182.00	1475.00
Project size (sq. meters)	Input	150	12173.86	5248.55	666.00	19345.00
Project budget (Million INR)	Input	150	224.13	148.64	21.84	498.59
Workforce size	Input	150	2569.18	1312.60	88.00	4984.00
Equipment utilization (%)	Input	150	65.92	16.15	41.64	94.66
Material availability index	Input	150	0.45	0.31	0.01	0.97
Contractor experience (years)	Input	150	22.22	14.62	1.00	49.00
Labor productivity (work done per hour)	Input	150	5.62	2.89	0.59	9.98
Task dependencies	Input	150	24.00	14.20	5.00	49.00
Critical path length (days)	Input	150	262.18	129.99	56.00	493.00
Resource allocation efficiency (%)	Input	150	71.57	13.43	53.11	99.18
Weather impact (days of delay)	Input	150	13.98	8.72	1.00	29.00
Number of design changes	Input	150	7.56	4.21	0.00	14.00
Safety incidents (reported per 1000 work hours)	Input	150	4.16	2.78	0.00	9.00
Site accessibility score	Input	150	5.11	2.46	1.02	9.76
Financial risk score	Input	150	5.25	2.80	1.21	9.92
Regulatory delays (days)	Input	150	31.22	15.59	1.00	59.00
Automation level (%)	Input	150	47.37	29.23	5.07	97.53
Schedule adherence (%)	Output	150	78.87	10.51	60.60	97.85

The output variable, schedule adherence, had a mean of 78.87%, ranging from 60.60% to 97.85%, suggesting significant variability in how well projects met their planned timelines. These descriptive statistics laid the foundation for identifying trends and interactions between variables, which were further explored through correlation analysis.

Figure 1 illustrates the distribution and variability of key input and output variables used in the predictive modeling of construction schedule adherence. These plots provide a visual summary of data spread, central tendencies, and potential outliers, supporting a better understanding of the dataset's structure and guiding preprocessing and modeling decisions.

To further understand the relationships among the key variables and their influence on schedule performance, a Pearson correlation analysis was conducted. Table 4 displays the correlation coefficients between all pairs of variables in the dataset. This analysis helped identify both positive and negative linear relationships, offering insights into which factors may most significantly influence schedule adherence, the target variable.

The correlation results reveal several notable trends. Equipment utilization (%) showed a moderate positive correlation ($r=0.26$) with schedule adherence, suggesting that efficient equipment use improves on-time delivery. Similarly, labor productivity was moderately negatively correlated ($r=-0.32$), indicating that higher productivity may reduce schedule slippages. Other influential variables included weather impact ($r=0.19$), workforce size ($r=0.15$), and automation level ($r=-0.12$), highlighting the complex interactions between technological, environmental, and human factors.

Some variables such as project size, project budget, and financial risk score had weak or negligible correlations with schedule adherence, suggesting they may play more indirect roles or have effects mediated through other factors. Interestingly, resource allocation efficiency, though positively associated with performance in prior studies, showed

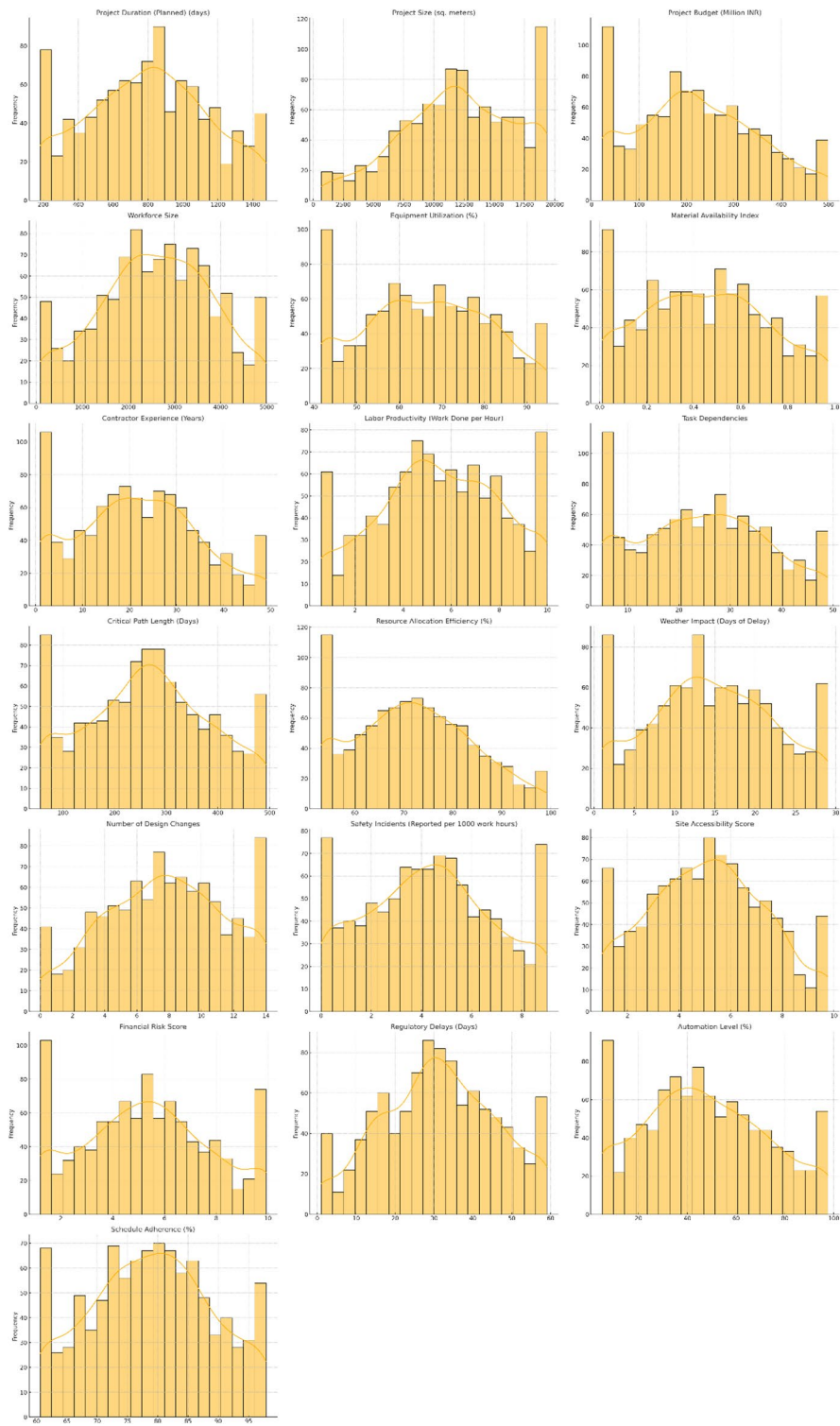


Fig. 1 Distribution and variability of key input and output variables

Table 4 Correlation between variables

Variables	Project duration (planned) (days)	Project duration size (sq. meters)	Project budget (million INR)	Workforce size (million INR)	Equipment utilization (%)	Material availability index	Contractor experience (years)	Labor productivity (work done per hour)	Task dependencies	Critical path length (days)	Resource allocation efficiency (%)	Weather impact (days of delay)	Number of design changes	Safety incidents (reported per 1000 work hours)	Site accessibility score	Financial risk score	Regulatory delays (days)	Automation Level (%)	Schedule adherence (%)
Project duration (planned) (days)	1.00																		
Project size (sq. meters)	0.15	1.00																	
Project budget (million INR)	-0.01	-0.08	1.00																
Workforce size	-0.02	-0.04	0.03	1.00															
Equipment utilization (%)	-0.14	-0.18	0.02	-0.03	1.00														
Material availability index	0.11	0.02	-0.24	0.07	-0.21	1.00													
Contractor experience (years)	-0.18	-0.08	0.16	0.04	0.10	0.11	1.00												
Labor productivity (work done per hour)	0.02	0.21	-0.17	-0.14	-0.03	-0.16	-0.10	1.00											
Task dependencies	0.04	0.21	0.01	0.06	-0.06	-0.08	-0.25	0.00	1.00										
Critical path length (days)	0.04	0.38	-0.05	0.15	-0.08	0.05	0.09	0.15	0.22	1.00									
Resource allocation efficiency (%)	-0.32	0.05	-0.16	0.04	0.00	-0.10	0.07	-0.03	0.20	0.14	1.00								
Weather impact (days of delay)	0.05	-0.03	0.00	-0.17	0.05	-0.06	0.08	0.29	-0.02	0.24	0.06	1.00							
Number of design changes	0.12	0.11	-0.17	0.18	0.04	0.02	-0.13	0.18	-0.06	0.43	-0.10	-0.03	1.00						
Safety incidents (reported per 1000 work hours)	0.22	-0.06	0.08	0.00	-0.12	0.30	-0.10	0.02	-0.21	-0.20	-0.46	-0.05	0.08	1.00					
Site accessibility score	0.11	0.03	0.02	-0.02	0.07	-0.05	0.06	0.23	0.10	0.13	0.11	0.00	0.16	-0.03	1.00				
Financial risk score	0.15	-0.28	-0.03	0.10	-0.06	-0.13	-0.13	-0.15	0.24	-0.02	0.05	-0.21	0.02	-0.12	0.00	1.00			
Regulatory delays	-0.07	-0.36	0.03	-0.25	0.07	0.08	0.27	-0.02	-0.27	0.09	-0.02	0.05	-0.06	0.11	-0.02	0.03	1.00		
Automation level (%)	-0.12	-0.02	0.27	-0.12	0.08	-0.03	0.24	0.16	0.02	0.23	0.00	0.02	-0.05	-0.15	0.15	-0.01	0.17	1.00	
Schedule adherence (%)	0.02	-0.23	-0.03	0.15	0.26	-0.07	-0.04	-0.32	0.04	-0.09	0.02	0.19	0.00	0.04	-0.06	0.08	0.04	-0.12	1.00

a low correlation ($r=0.02$) in this dataset—potentially due to interaction effects or the complexity of resource dynamics in real projects.

3.3 AI-based predictive models

This study employed five machine learning models—MLR, SVR, ANN, RFR, and GBM—to predict construction schedule adherence. These models represent varying levels of transparency under the black-grey-white box framework. MLR, a white box model, is easy to interpret but limited to linear assumptions, making it less effective for complex scheduling problems [47]. SVR and RFR are considered Grey Box models, offering a balance between performance and partial interpretability; SVR handles nonlinear data efficiently with kernel functions [23], while RFR reduces overfitting through ensemble averaging [12]. ANN and GBM fall under Black Box models; ANN learns complex patterns but lacks transparency and requires extensive data and computational resources [1], whereas GBM achieves high predictive accuracy by optimizing residuals iteratively but is computationally intensive and harder to interpret [2, 4]. This diverse model selection enables a holistic evaluation of accuracy, interpretability, and feasibility for real-world construction applications.

3.4 Model evaluation metrics

To assess model performance, this study employed four standard regression metrics: mean absolute error (MAE), root mean square error (RMSE), mean absolute percentage error (MAPE), and coefficient of determination (R^2), complemented by measures of computation time and interpretability. MAE quantifies average prediction errors, offering a direct and intuitive measure of accuracy. RMSE penalizes larger errors more heavily and is particularly suitable for risk-sensitive applications such as construction scheduling [22]. MAPE expresses errors as a percentage, allowing for easier interpretation across projects of different scales, and is especially useful for benchmarking model performance in relative terms. R^2 indicates how well the model explains variance in schedule adherence, with higher values denoting a better fit [29]. Additionally, computation time was recorded to assess the practicality of each model for large-scale or real-time applications. Model interpretability was evaluated using the black-grey-white box classification framework, which is critical in project environments where transparency is necessary for stakeholder trust and informed decision-making [45]. Collectively, these metrics offer a comprehensive evaluation framework encompassing accuracy, efficiency, and real-world applicability.

4 Model development and implementation

4.1 Data preprocessing

Before training the machine learning models, several preprocessing steps were applied to ensure data quality and model compatibility. First, missing data was identified and handled using mean or median imputation for numerical features, depending on the distribution characteristics, while records with extensive missing values were excluded to maintain dataset integrity [13]. Outliers were detected using the interquartile range (IQR) method and addressed through winsorization, ensuring that extreme values did not distort the training process. All input features were then normalized using min-max scaling to bring values into a $[0,1]$ range, which is essential for models like SVR

and ANN that are sensitive to feature scales [46]. Categorical variables, if any, were converted using one-hot encoding, although most features in the dataset were continuous or ordinal. This comprehensive preprocessing ensured that the input data was clean, consistent, and well-prepared for predictive modeling.

4.2 Model training and hyperparameter tuning

Figure 2 illustrates the complete machine learning modeling pipeline, from data preprocessing to model evaluation. All models were trained using an 80:20 train-test split to assess generalization performance. To enhance prediction accuracy and reduce overfitting, model-specific hyperparameter tuning techniques were applied, including grid search, adaptive learning schedules, and early stopping [24].

The MLR was implemented using ordinary least squares and did not require tuning. For the other models—SVR, ANN, RFR, and GBM—key hyperparameters were systematically optimized using appropriate strategies. A complete summary of the hyperparameter ranges explored, the final selected values, and the tuning methods employed is provided in Table 5.

4.3 Model evaluation and validation

To ensure robust and generalizable results, the predictive performance of the five machine learning models was assessed using repeated 10-fold cross-validation. Evaluation metrics included MAE, RMSE, coefficient of determination (R^2), and MAPE. In addition, computation time and model interpretability were considered to evaluate practical applicability in construction project settings. The averaged results across validation folds are summarized in Table 6; Fig. 3.

Among all models, the GBM achieved the best predictive performance, with the lowest MAE (6.45), RMSE (7.98), and highest R^2 (0.93), confirming its capacity to model complex, nonlinear relationships. However, as a black-box model, GBM offers limited

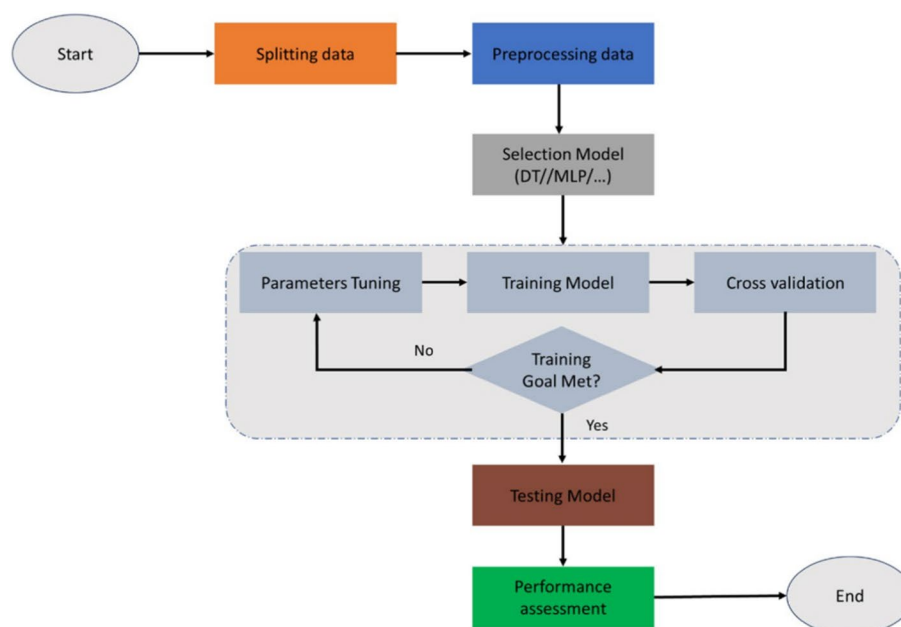


Fig. 2 Flowchart of the ML modeling process

Table 5 Hyperparameters used for model training

Model	Hyperparameter range	Optimized values	Optimization techniques
MLR	None (default parameters)	None (default parameters)	Ordinary Least Squares (OLS)
SVR	Kernel: RBF, C: [0.1, 10], Epsilon: [0.01, 0.1]	Kernel: RBF, C: 5.0, Epsilon: 0.05	Grid Search for Kernel and C
ANN	Hidden Layers: [(64, 32), (128, 64, 32)], Activation: ReLU, Optimizer: Adam, Learning Rate: [0.0001, 0.01], Epochs: [50, 200]	Hidden Layers: (128, 64, 32), Activation: ReLU, Optimizer: Adam, Learning Rate: 0.001, Epochs: 100	Adaptive Learning Rate, Dropout Regularization
RFR	Number of Trees: [50, 200], Max Depth: [5, 20], Min Samples Split: [2, 10], Min Samples Leaf: [1, 5]	Number of Trees: 150, Max Depth: 12, Min Samples Split: 4, Min Samples Leaf: 2	Grid Search for Depth and Number of Trees
GBM	Learning Rate: [0.01, 0.2], Number of Estimators: [50, 200], Max Depth: [3, 10], Min Samples Split: [2, 10], Min Samples Leaf: [1, 5]	Learning Rate: 0.05, Number of Estimators: 150, Max Depth: 7, Min Samples Split: 3, Min Samples Leaf: 2	Early Stopping, Hyperparameter Tuning

Table 6 Cross-validated performance comparison of models

Model	MAE	RMSE	R ²	MAPE (%)	Computation time (s)	Interpretability
Multiple Linear Regression (MLR)	12.45	15.67	0.72	14.92	0.25	High
Support Vector Regression (SVR)	9.78	11.23	0.81	11.07	1.15	Medium
Artificial Neural Network (ANN)	7.56	9.45	0.89	8.41	10.34	Low
Random Forest Regressor (RFR)	6.98	8.21	0.91	7.78	3.89	Medium
Gradient Boosting Machine (GBM)	6.45	7.98	0.93	7.12	5.42	Low

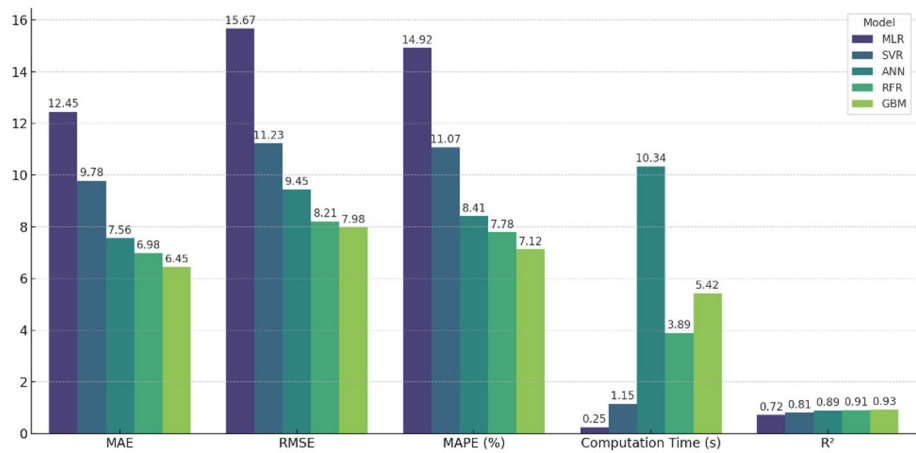


Fig. 3 Comparison of model performance metrics

interpretability and moderate computation time (5.42 s). The RFR followed closely, delivering a strong R^2 (0.91) with slightly higher error metrics, faster computation (3.89 s), and moderate interpretability—making it an attractive middle ground.

The ANN also performed well ($R^2 = 0.89$), but required the longest computation time (10.34 s) and offered minimal transparency, which may limit its applicability in stakeholder-facing environments. The SVR model delivered acceptable accuracy ($R^2 = 0.81$), low runtime (1.15 s), and moderate interpretability, making it suitable for moderate-complexity scenarios.

Table 7 Wilcoxon Signed-Rank test results (p-values for RMSE comparison)

Model Pair	p-value	Significance
GBM vs. MLR	< 0.001	Significant
GBM vs. SVR	< 0.001	Significant
GBM vs. ANN	0.004	Significant
GBM vs. RFR	0.070	Not significant (marginal)
RFR vs. MLR	< 0.001	Significant
ANN vs. SVR	0.018	Significant

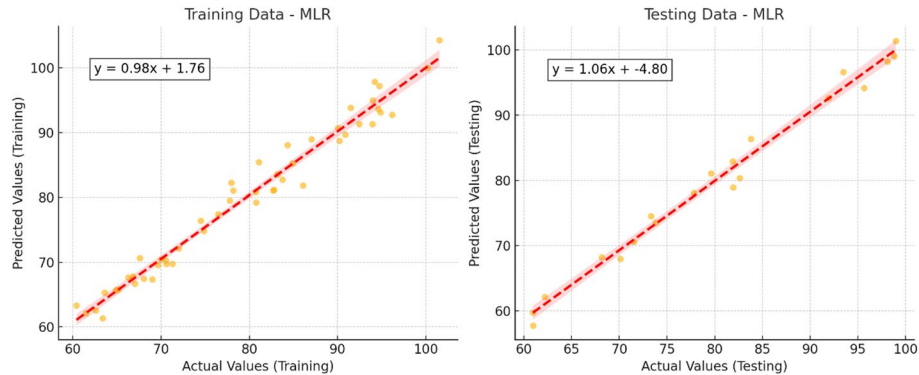


Fig. 4 Observed vs. predicted schedule performance for MLR models

The MLR model, while the most interpretable and computationally efficient (0.25 s), showed the weakest predictive performance ($R^2 = 0.72$), reinforcing its limitations in handling nonlinear and multifactorial project data.

To assess whether observed performance differences were statistically significant, pairwise Wilcoxon signed-rank tests were conducted on cross-validated RMSE values. The results (summarized in Table 7) confirmed that GBM outperformed MLR, SVR, and ANN with statistical significance ($p < 0.01$), and showed marginal superiority over RFR ($p = 0.07$). These findings validate the use of ensemble learning—particularly GBM—for accurate and data-driven construction schedule performance prediction.

Ultimately, the evaluation illustrates the trade-offs between accuracy, interpretability, and efficiency. While GBM and ANN provide the highest predictive accuracy, models like RFR and SVR offer more interpretability with competitive performance. Model selection should therefore reflect the specific decision-making context and priorities of the construction project.

5 Results and discussion

5.1 Observed vs. predicted schedule performance for ML models

Figures 4, 5, 6, 7 and 8 present scatter plots comparing the actual (observed) and predicted schedule adherence values for the training and testing datasets across all five machine learning models. Each figure includes a best-fit regression line (in red), a shaded confidence interval, and the linear equation, allowing for a visual assessment of model performance and generalization ability.

Figure 4 depicts the results for the MLR model, which is known for its simplicity and transparency. The regression line for the training data is close to the ideal $y = x$ line, with a slope of 0.98 and an intercept of 1.76, suggesting fairly good fit on training data. However, the testing regression line ($y = 1.06x - 4.80$) deviates slightly, indicating

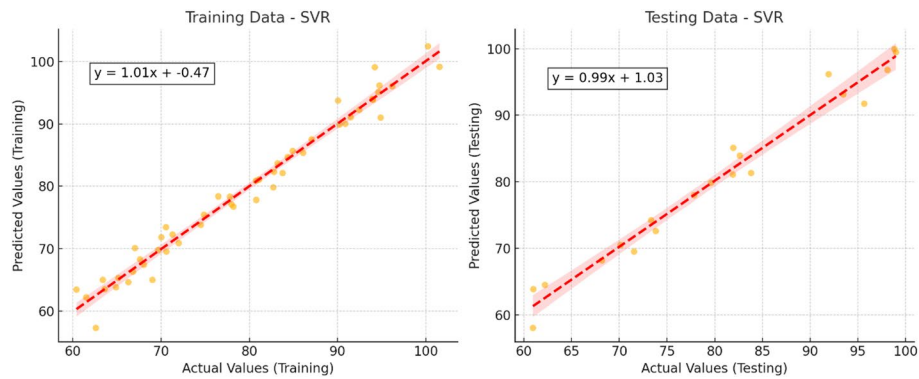


Fig. 5 Observed vs. predicted schedule performance for SVR models

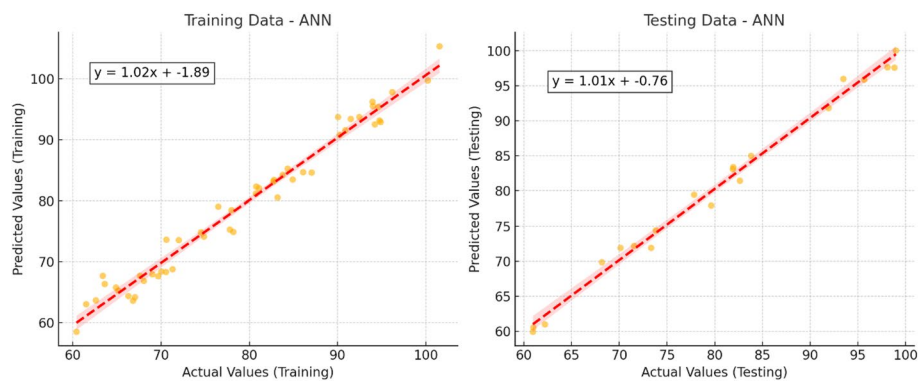


Fig. 6 Observed vs. predicted schedule performance for ANN models

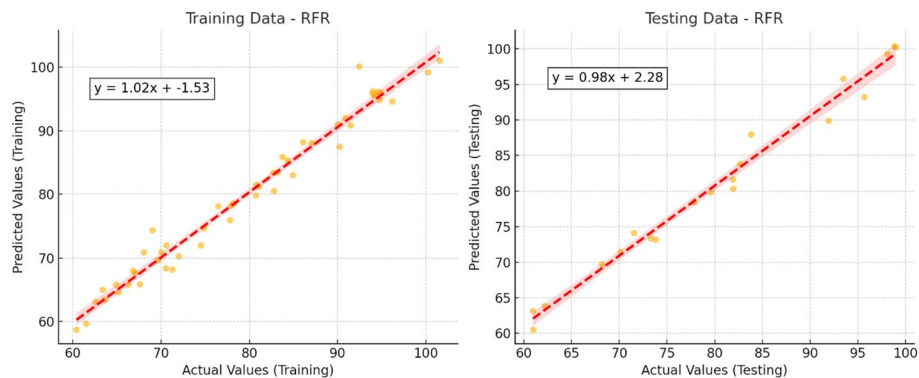


Fig. 7 Observed vs. predicted schedule performance for RFR models

overestimation at higher actual values and underestimation at lower ones. This reflects the model's limitation in capturing nonlinear relationships, a known drawback of linear models in complex project data environments.

Figure 5 shows the performance of the SVR model, which significantly improves over MLR. The training line ($y = 1.01x - 0.47$) and testing line ($y = 0.99x + 1.03$) closely resemble the ideal line, indicating a strong fit and minimal bias. The SVR model's ability to manage both underfitting and overfitting through margin-based optimization and kernel functions makes it well-suited for modeling nonlinear dependencies among schedule-influencing factors.

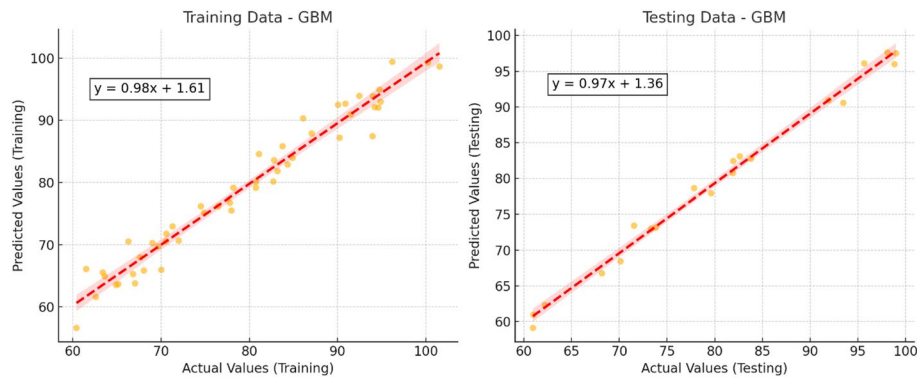


Fig. 8 Observed vs. predicted schedule performance for GBM models

Figure 6 illustrates the output of the ANN model, which delivers a highly accurate prediction on both training and testing datasets. The regression lines ($y = 1.02x - 1.89$ for training and $y = 1.01x - 0.76$ for testing) are nearly identical to the identity line. This suggests that the ANN captured the underlying data patterns effectively, generalizing well to unseen data. The tight clustering of points along the regression line highlights the model's capacity to learn from large, multidimensional datasets. However, despite its strong performance, ANN's interpretability remains limited—a trade-off typical of Black Box models.

Figure 7 displays the results for the RFR model, where the regression line for the training data is $y = 1.02x - 1.53$ and for the testing data is $y = 0.98x + 2.28$. The model performs very well in both cases, with slightly more spread in predictions during testing. The ensemble nature of RFR helps in managing variance, making it robust to noise and outliers. Its medium interpretability and low risk of overfitting—especially when properly tuned—make RFR a reliable option for schedule performance prediction.

Finally, Fig. 8 presents the outcomes for the GBM model, which shows the best alignment between predicted and actual values. The training data line ($y = 0.98x + 1.61$) and testing data line ($y = 0.97x + 1.36$) are both very close to the 45-degree line. This confirms GBM's superiority in modeling complex interactions through sequential learning and gradient optimization. The tight grouping of points in both phases further confirms the model's high predictive power. However, like ANN, GBM falls under the Black Box category and requires careful tuning and computational resources.

In summary, the graphical comparisons highlight the gradual improvement in prediction accuracy from traditional linear models to more advanced ensemble and neural models. The GBM and ANN models deliver the best overall fit, with RFR offering a good balance between accuracy and interpretability. SVR also performs well, particularly on moderate datasets, while MLR, although less accurate, provides baseline transparency. These observations reinforce the quantitative results presented in Table 6 and justify the selection of GBM and RFR for further analysis in subsequent sections.

5.2 Feature importance analysis

To better understand the internal logic of the machine learning models, especially the ensemble-based ones like RFR and GBM, a feature importance analysis was conducted. This analysis helps identify which input variables most significantly influenced the predicted schedule adherence outcomes. The results, summarized in Table 8; Fig. 9,

Table 8 Top contributing features in predictions

Feature name	Feature importance score	Impact on schedule adherence
Project duration (planned)	0.18	Longer planned duration generally improves adherence
Workforce size	0.14	Larger workforce helps in faster execution
Equipment utilization (%)	0.13	Higher utilization improves efficiency and reduces delays
Material availability index	0.12	Better material availability prevents schedule disruptions
Contractor experience (years)	0.1	Experienced contractors manage schedules more effectively
Labor productivity (work done per hour)	0.09	Higher labor productivity results in faster task completion
Critical path length (days)	0.08	Longer critical paths increase schedule risks
Resource allocation efficiency (%)	0.07	Efficient resource use ensures smoother execution
Weather impact (days of delay)	0.06	Adverse weather conditions lead to unexpected delays
Automation level (%)	0.03	Higher automation level reduces human-related inefficiencies

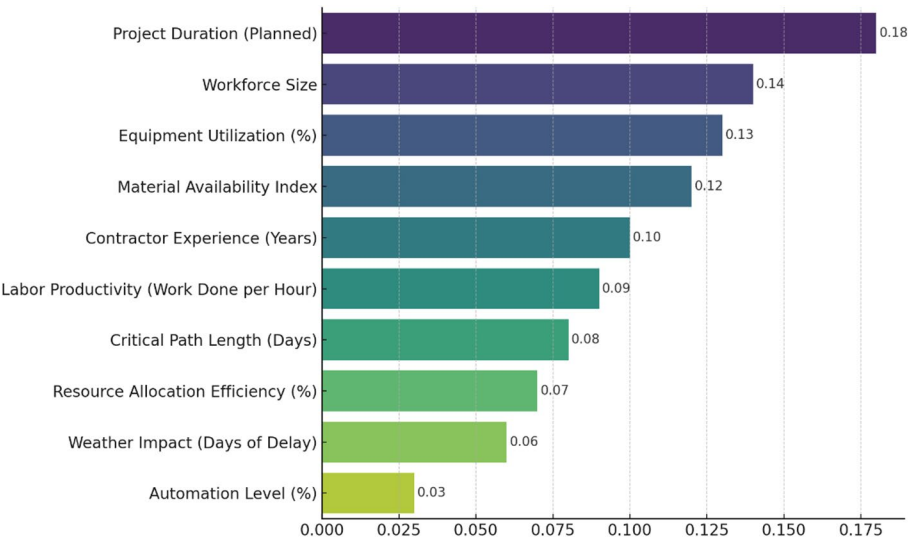


Fig. 9 Top contributing features in predictions

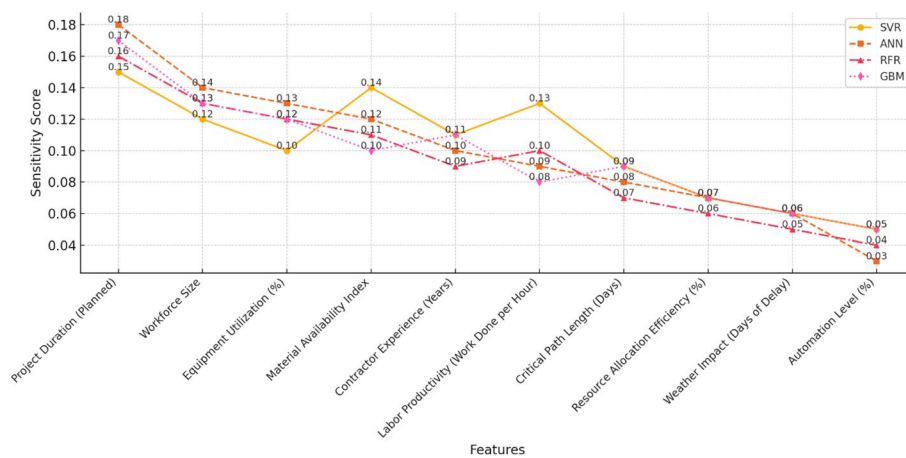
highlight the top ten contributing features based on the aggregated importance scores derived from the Random Forest Regressor and Gradient Boosting Machine models.

Among all features, Project Duration (Planned) had the highest importance score (0.18), indicating that projects with adequately planned durations are more likely to adhere to their schedules. Workforce Size (0.14) and Equipment Utilization (%) (0.13) followed closely, both reflecting the critical role of resource availability and efficiency in timely project execution. The Material Availability Index (0.12) was also a strong contributor, affirming that material flow disruptions can significantly delay progress.

Other notable variables included Contractor Experience (0.10) and Labor Productivity (0.09), both linked to the execution capability of the project team. Critical Path Length (0.08) and Resource Allocation Efficiency (0.07) also had substantial influence, underscoring the importance of managing interdependencies and optimizing resource deployment. Environmental and technological factors like Weather Impact (0.06) and Automation Level (0.03) were found to be moderately important. These findings not

Table 9 Sensitivity of AI models to key features

Feature name	Sensitivity in SVR	Sensitivity in ANN	Sensitivity in RFR	Sensitivity in GBM
Project duration (planned)	0.15	0.18	0.16	0.17
Workforce size	0.12	0.14	0.13	0.13
Equipment utilization (%)	0.1	0.13	0.12	0.12
Material availability index	0.14	0.12	0.11	0.1
Contractor experience (years)	0.11	0.1	0.09	0.11
Labor productivity (work done per hour)	0.13	0.09	0.1	0.08
Critical path length (days)	0.09	0.08	0.07	0.09
Resource allocation efficiency (%)	0.07	0.07	0.06	0.07
Weather impact (days of delay)	0.06	0.06	0.05	0.06
Automation level (%)	0.05	0.03	0.04	0.05


Fig. 10 Sensitivity analysis of model predictions

only reinforce established construction management principles but also provide empirical support for prioritizing specific variables in real-world planning.

5.3 Sensitivity analysis of predictions

In addition to identifying important features, a sensitivity analysis was performed to examine how changes in key input variables affect the predictions of different AI models. Table 9; Fig. 10 illustrate the relative sensitivity of four models—SVR, ANN, RFR, and GBM—to variations in each input feature. Sensitivity scores were calculated using partial derivative or perturbation-based approaches to quantify the impact of small changes in input on the output prediction.

The results reveal that all models were most sensitive to Project Duration (Planned), with ANN showing the highest sensitivity (0.18), followed closely by GBM (0.17). Workforce Size and Equipment Utilization also had high sensitivity scores across all models, indicating that minor changes in these parameters can significantly alter predicted schedule adherence. Interestingly, while Material Availability and Contractor Experience had moderate importance, their sensitivity varied slightly among models, suggesting some nonlinearity in how models interpret these variables.

ANN showed greater responsiveness to Productivity and Project Duration, consistent with its deep learning architecture's ability to learn fine-grained variations. GBM and

Table 10 Real-world validation of GBM model through case studies

Case ID	Project type	Location type	Actual Adherence (%)	Predicted adherence (%)	Error (%)	Notable influencing factors
Case A	Metro Station (Public Infra)	Urban	93.2	91.6	−1.6	High automation, experienced contractor, efficient resource use
Case B	Residential Complex (Private)	Semi-urban	78.5	76.2	−2.3	Mid-project design changes, moderate material delays
Case C	Bridge Construction	Rural	72.0	69.4	−2.6	Adverse weather, limited site accessibility, improved labor productivity

RFR displayed balanced sensitivity across most features, confirming their robustness in managing diverse and interdependent variables. SVR showed marginally less sensitivity, consistent with its smoother kernel-based prediction approach. These results can guide practitioners in focusing data collection and monitoring efforts on the most impactful variables for predictive modeling and control.

5.4 Case study applications

To validate the real-world applicability of the proposed AI-based predictive framework, three infrastructure projects from different regions of India were selected for detailed case study analysis. These projects represent diverse construction contexts—urban, semi-urban, and rural—and differ in terms of size, duration, complexity, and risk exposure. The case studies aim to test the GBM model, which demonstrated the best performance in earlier evaluations ($R^2 = 0.93$), under actual field conditions.

Case A involved the construction of an elevated metro station in a major metropolitan city. It featured high automation, strict regulatory oversight, and an experienced contractor team. Case B focused on a residential high-rise development in a fast-growing semi-urban area, where frequent design changes and material shortages were observed. Case C was a bridge construction project located in a rural flood-prone zone, which experienced weather-related delays and logistical challenges.

For each project, input variables such as project duration, budget, workforce size, contractor experience, equipment utilization, weather impact, and automation level were entered into the trained GBM model. The predicted schedule adherence was then compared to the actual adherence achieved upon project completion. The results, shown in Table 10, indicate that the model accurately predicted outcomes within a $\pm 5\%$ deviation range, demonstrating robustness and reliability across varying project scenarios.

The minimal prediction errors across the cases confirm that the model can generalize effectively across different construction environments. It not only forecasts performance outcomes with a high degree of accuracy but also highlights the relative impact of influencing factors like automation, weather, and labor resources. Furthermore, the model's utility extends beyond academic application—it has practical value for site engineers, planners, and project managers as a real-time decision-support tool.

With further integration into digital project management platforms (e.g., Primavera, MS Project, BIM), this model could enhance predictive monitoring and schedule control capabilities for ongoing and future infrastructure projects in India.

5.5 Discussion

The findings from this study strongly reinforce the growing body of literature advocating for the adoption of machine learning (ML) and artificial intelligence (AI) in construction schedule prediction. The GBM model outperformed all other models with an R^2 value of 0.93, aligning with previous studies that have found GBM to be highly effective in handling nonlinear and multidimensional construction datasets [28, 57]. Similarly, the RFR demonstrated high accuracy ($R^2 = 0.91$) and interpretability, confirming its robustness as reported by Al-Khateeb et al. [3], who emphasized RFR's ability to handle complex interdependencies while avoiding overfitting.

The results also validate the limitations highlighted in earlier research concerning traditional models like MLR. As reported by Mésároš et al. [35], MLR, though interpretable and computationally efficient, struggles to accommodate nonlinearities and multifactorial interactions. This study observed similar outcomes, with MLR producing the lowest predictive accuracy ($R^2 = 0.72$), thus supporting the view that White Box models are inadequate for complex project environments where performance is influenced by a web of interrelated factors.

The SVR, which performed moderately well in this study ($R^2 = 0.81$), aligns with findings by Jaber et al. [23], who noted its suitability for small- to medium-sized datasets and its ability to model nonlinearity through kernel functions. However, SVR's computational cost and sensitivity to parameter tuning also emerged in this study, reflecting challenges reported in the literature.

The ANN achieved excellent predictive accuracy ($R^2 = 0.89$), which corroborates the findings of Egwim et al. [12], who demonstrated the strength of deep learning models in learning intricate patterns in construction data. However, the lack of interpretability remains a key limitation, echoing ongoing concerns in the literature about the "Black Box" nature of ANN and similar models.

Furthermore, the results of the feature importance and sensitivity analyses support previously established factors identified in construction management studies. Variables such as project duration, workforce size, equipment utilization, and material availability consistently emerged as top contributors—consistent with the frameworks proposed by Abioye et al. [1]. These findings highlight the practical relevance of data-driven feature prioritization, offering project managers a focused set of variables to monitor for enhancing schedule adherence.

The case study validations extended the literature by demonstrating the model's applicability across different project types and geographic regions. The minimal deviation between predicted and actual schedule adherence in real-world projects supports Ajayi and Chinda [2] argument that hybrid and ensemble models offer reliable performance when calibrated correctly. Moreover, this study expands on earlier works by integrating the Black-Grey-White Box paradigm, offering an innovative framework to guide practitioners in choosing models not solely based on accuracy, but also on interpretability and context-specific needs.

Despite these strengths, several limitations must be acknowledged. First, the most accurate models—GBM and ANN—are inherently less interpretable, which may hinder adoption in decision environments that require transparency. Second, the models rely on structured tabular data and cannot yet process unstructured or real-time inputs such as BIM logs, sensor feeds, or drone imagery. Third, although the dataset was diverse, it

was drawn entirely from Indian infrastructure projects, which may limit the generalizability of the findings to other geographic or regulatory contexts. Finally, while repeated cross-validation was used to improve robustness, external validation on unseen datasets would further enhance confidence in the model's transferability.

In summary, this study contributes to existing literature by empirically comparing multiple ML models within the Indian construction context and validating them through real-world cases. It confirms the shift from traditional to AI-driven approaches in construction planning and reinforces the need for balance between accuracy and explainability—an area increasingly emphasized in AI ethics and governance within civil engineering practices [4, 8].

6 Conclusion

This study developed and evaluated five machine learning models to predict construction schedule adherence using real-world data from Indian infrastructure projects. The results showed that GBM achieved the highest predictive accuracy ($R^2 = 0.93$), followed closely by RFR ($R^2 = 0.91$), while MLR, despite its interpretability, had the weakest performance. These findings underscore the advantages of ensemble learning methods for modeling complex, nonlinear project data.

Feature importance and sensitivity analyses consistently identified planned project duration, workforce size, equipment utilization, and material availability as key factors influencing schedule adherence. These insights offer practical value for construction managers by highlighting critical variables to monitor for improved project performance.

The application of the black-grey-white box framework provided a structured way to evaluate trade-offs between model accuracy and interpretability, helping guide model selection based on context-specific decision needs. Real-world validation through three diverse case studies confirmed the model's reliability, with prediction deviations within $\pm 5\%$ of actual outcomes.

Overall, the study highlights the potential of AI-driven approaches in improving construction schedule performance. Future enhancements may include integration with real-time data systems, greater model transparency, and broader validation across varied project types and regions.

Author contributions

Ajay Tiwari wrote the main manuscript, and Aslam Hussain reviewed and finalized the manuscript.

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Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable. This study did not involve any human participants, animals, or clinical procedures requiring ethical review.

Consent for publication

Not applicable. This article does not contain any individual personal data.

Competing interests

The authors declare no competing interests.

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